

Amina Rakhimbergenova

amina@uni.minerva.edu | [GitHub](#) | [LinkedIn](#) | aminarhe.me | San Francisco, CA

EDUCATION

Minerva University

B.S. in Computational Sciences (Mathematics and Artificial Intelligence)

Expected May 2027

San Francisco, CA

Extracurriculars: Bio-inspired Robotics Club, Mathematical Modeling Research Innovation Group, AI Safety.

Relevant Coursework: Machine Learning, Optimization, Statistics, Linear Algebra, Data Structures, Algorithms.

TECHNICAL SKILLS

Languages: C++, Python, TypeScript, Bash, SQL.

Robotics & Perception: Model Predictive Control, MuJoCo, OpenCV, Google OR-Tools, KiCAD, Onshape CAD, PCB design and bring-up, oscilloscope bench debugging.

Systems: Linux, Docker, Kubernetes, Slurm, FastAPI, GitHub Actions, CI/CD, PyTorch.

EXPERIENCE

Infrastructure & Software Engineer Intern

May 2026 – Aug 2026

Lucid Computing

San Francisco, CA

- Built heterogeneous 16-GPU cluster (8× NVIDIA H100, 8× A100) under one Slurm-on-Kubernetes scheduler; made deployment reproducible via GitOps/IaC, spinning up separate dev cluster from same declarative config, no manual setup.
- Shipped the researcher-facing platform: ~1,900-line FastAPI control plane as the cluster's single provisioning and audit path (idempotent identity, storage, compute quota, and usage accounting operations across Slurm, LDAP, NFS, and Kubernetes), and a React console for self-serve GPU provisioning, SSH access, and role-based access.
- Deployed distributed LLM inference, served Qwen3.5-397B-A17B tensor-parallel across 8×H100 GPUs, maintained as code and guarded by CI smoke test that distinguishes real regressions from busy-cluster states.
- Set up CI/CD system with tag-triggered automated deploys and rolling updates on self-hosted runners inside air-gapped cluster, with rollout verification and version pinning; added self-hosted LLM-based PR review gate before production.

Robotics Engineer Intern

Jan 2026 – Mar 2026

Trilobio

San Francisco, CA

- Developed Model Predictive Control (MPC) optimization framework for robotic arm motion planning, eliminating overshoot and achieving ~2× faster settling and 96–99.7% lower steady-state error.
- Stood up MuJoCo physics simulation environment for hardware-in-the-loop validation of robotic arm control algorithms, enabling rapid controller iteration before hardware deployment.
- Benchmarked latency, stability, and computational load to assess feasibility for real-time MPC embedded robotic deployment.

Software Engineer Intern

June 2025 – Aug 2025

Trilobio

San Francisco, CA

- Built multichannel & multidispense pipetting optimization system in TypeScript with transformation matrix-based rotation planning for robot arm; enabled parallel operations that sped up lab protocols by 6x.
- Designed contamination-modeling algorithm to prevent cross-contamination; reduced consumable waste by 30%.
- Architected auto-layout optimization engine using constraint programming with Python, Google OR-Tools, custom solvers, to automate lab protocol planning; replaced hours of biologists' daily manual work.

Robotics Engineer Intern

May 2024 – Aug 2024

Trilobio

San Francisco, CA

- Re-architected the robot communication stack from CAN (8 Mbps) to USB 2.0 High-Speed (480 Mbps), validating signal integrity with an oscilloscope and test fixtures.
- Engineered and debugged custom USB hub PCBs and firmware, enabling high-throughput sensor streaming.
- Designed CAD camera and laser mounting assemblies and developed OpenCV-based laser-triangulation pipeline to generate 3D point clouds, and surface mesh models for robotic perception.

PROJECTS

MARS Robot Sous Chef (RoboHacks) | *Python, ACT, Gemini, ElevenLabs, Robotics*

- Developed multimodal embodied AI agent on MARS Innate robotic platform using Gemini for visual verification and task planning and ElevenLabs API for voice chat.
- Trained ACT imitation-learning policy enabling robot to perform manipulation tasks including cutlery operation.

Sim-Francisco (NVIDIA Spark Hack) | *Python, RAPIDS (cuDF, cuSpatial, CuPy), GPU Computing*

- Built GPU-accelerated city-scale multi-agent earthquake evacuation simulation using NVIDIA RAPIDS, optimizing memory locality and enabling real-time pathfinding across large geospatial datasets on unified GPU memory.
- Modeled seismic risk, debris propagation, and multi-agent movement using USGS and San Francisco geospatial datasets.